

## Key Competencies & Knowledge Areas

- **Programming Languages:** JavaScript (core: Node.js + React), Python (backend), HTML, CSS
- **Version Control:** Git & GitHub
- **Frontend Development:** HTML5, CSS3, JavaScript, Responsive Design, React (or Angular/Vue)
- **Backend Development:** Node.js (Express), Python (Flask/Django), API design (REST, GraphQL)
- **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB), ORM tools
- **Deployment & DevOps:** Fundamentals of cloud hosting (AWS/Heroku/Vercel), CI/CD basics
- **Testing:** Unit, integration, and end-to-end testing (Jest, Mocha, Pytest)
- **Software Architecture:** MVC, microservices, design patterns
- **Soft Skills:** Documentation, code reviews, agile development practices

## Sequenced Learning Milestones

### Short-Term (Months 1–3): Foundations

#### 1. Programming Basics & Git

- **Resources:**
  - [freeCodeCamp JavaScript](#)
  - [Codecademy Python Course](#)
  - [GitHub Docs](#)
- **Project:** Build a simple calculator & version control with Git.

#### 2. Frontend Fundamentals: HTML, CSS, JavaScript

- **Resources:**
  - [MDN Web Docs](#)
  - [freeCodeCamp Responsive Web Design](#)
- **Project:** Personal portfolio webpage (responsive design).

### 3. Version Control & Collaboration

- **Resources:**
  - [Git and GitHub Crash Course \(YouTube\)](#)
  - [Git Branching Practice](#)
- **Project:** Host your project on GitHub, practice branching & pull requests.

## Mid-Term (Months 4–7): Fullstack Foundations

### 4. Advanced Frontend (React or another framework)

- **Resources:**
  - [React Official Docs](#)
  - [Scrimba React Course](#)
- **Project:** Build a task manager SPA (single-page application).

### 5. Backend Fundamentals (Node.js/Express, Python/Flask)

- **Resources:**
  - [Node.js & Express \(MDN\)](#)
  - [Flask Mega-Tutorial](#)
- **Project:** RESTful API for your task manager app (CRUD operations).

### 6. Databases (SQL & NoSQL)

- **Resources:**
  - [SQL for Beginners \(Khan Academy\)](#)
  - [MongoDB Basics](#)
- **Project:** Integrate a PostgreSQL and/or MongoDB database into your backend project.

## Long-Term (Months 8–12): Advanced Fullstack & DevOps

### 7. Connecting Frontend to Backend (API Integration)

- **Resources:**
  - [Axios Docs \(Frontend -> API\)](#)
  - [React + Express Integration Guide \(Medium\)](#)
- **Project:** Deploy a full React + Express web app.

## 8. Testing & Debugging

- **Resources:**
  - [Jest \(JavaScript Testing\)](#)
  - [Pytest \(Python Testing\)](#)
- **Project:** Write unit and integration tests for previous projects.

## 9. Deployment & DevOps Basics

- **Resources:**
  - [Heroku Deployment Guide](#)
  - [Vercel for Frontend](#)
  - [CI/CD Basics \(Codecademy\)](#)
- **Project:** Deploy your fullstack app to Heroku or Vercel, set up automated tests on commit.

## 10. Software Architecture & Advanced Topics

- **Resources:**
  - [Patterns of Enterprise Application Architecture \(Book\)](#)
  - [Microservices Explained \(AWS\)](#)
- **Project:** Refactor a feature into a separate microservice, add thorough documentation.

## 11. Portfolio & Professional Growth

- **Resources:**
  - [Build a Github Portfolio](#)
- **Project:** Organize, document, and showcase all completed software projects.

## Step-by-Step Roadmap Structure

## **Step 1: Core Language & Git Mastery**

- Complete beginner JS/Python courses, daily small exercises.
- Initiate and manage all projects with Git.

## **Step 2: Build Frontend Projects**

- Create static and, later, dynamic web pages.
- Practice responsive design and accessibility.

## **Step 3: Backend Development**

- Build RESTful APIs, handle authentication, connect with databases.

## **Step 4: Database Integration**

- Use SQL and NoSQL models to store/query data.
- Connect backend projects to databases; work with relational and document-based designs.

## **Step 5: Fullstack Integration**

- Merge frontend and backend – interface with APIs.
- Handle state/data flows from user to database and back.

## **Step 6: Testing & DevOps**

- Write and run unit and integration tests.
- Learn basics of CI/CD and cloud deployment.

## **Step 7: Advanced Architecture & Specialization**

- Experiment with microservices, caching, scalable code.
- Write thorough documentation, contribute to open source.

## **Step 8: Portfolio Expansion**

- Document case studies for each major project.
- Update LinkedIn, GitHub, and project website.

### **Short-Term Goals (Months 1–6):**

- Launch 2–3 frontend and backend mini-projects.
- Master JavaScript, Python, and basic Git.
- Deploy first static website and API.

### **Long-Term Goals (Months 7–12):**

- Build and deploy a fully integrated fullstack app.
- Refactor at least one project for microservice architecture.
- Document and present all work in portfolio, prep for interviews.

### **Progress Tracking:**

- Weekly: Log coding hours, completed lessons, and new skills.
- Monthly: Finish and publish projects, review and debug code.
- Quarterly: Update portfolio, receive peer feedback, apply for higher-level development tasks.

**Following this roadmap** will help you develop practical and professional fullstack software development skills, leading toward career mobility and project confidence.